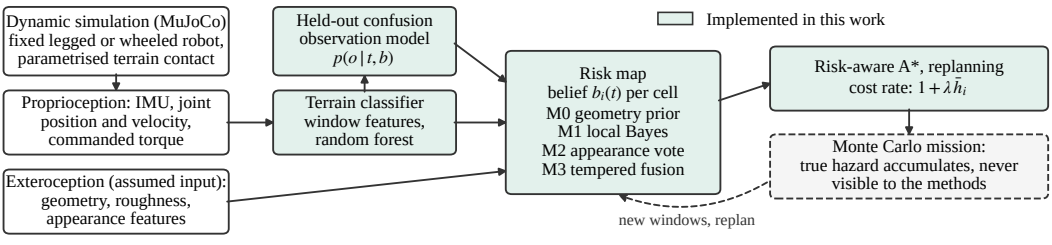
Dynamic simulation (MuJoCo)
fixed legged or wheeled robot,
parametrised terrain contact
Proprioception: IMU, joint
position and velocity,
commanded torque
Exteroception (assumed input):
geometry, roughness,
appearance features
Held-out confusion
observation model
$p(o \mid t, b)$
Terrain classifier
window features,
random forest
Implemented in this work
Risk map
belief $b_i(t)$ per cell
M0 geometry prior
M1 local Bayes
M2 appearance vote
M3 tempered fusion
Risk-aware A*, replanning
cost rate: $1 + \lambda \bar{h}_i$
Monte Carlo mission:
true hazard accumulates, never
visible to the methods
new windows, replan